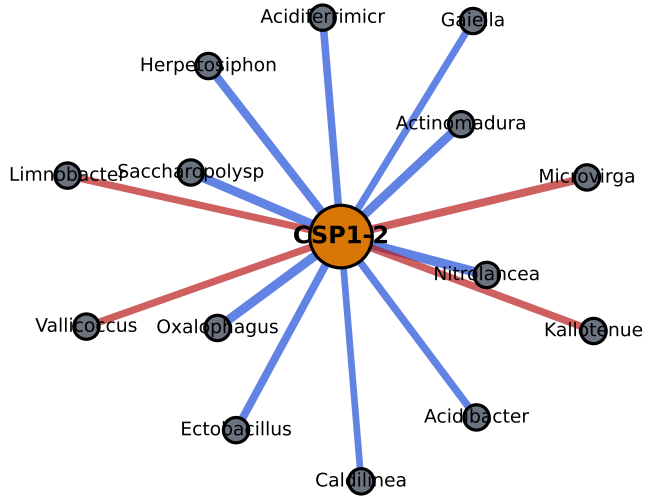
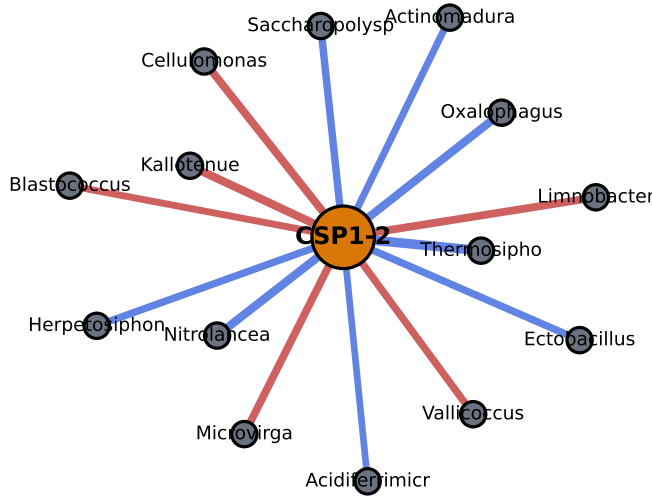


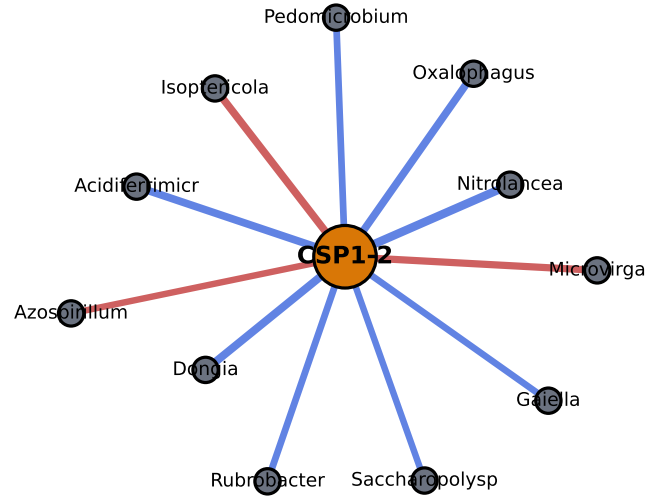
(a) Surface — CSP1-2 ego network (14 co-occurring genera)



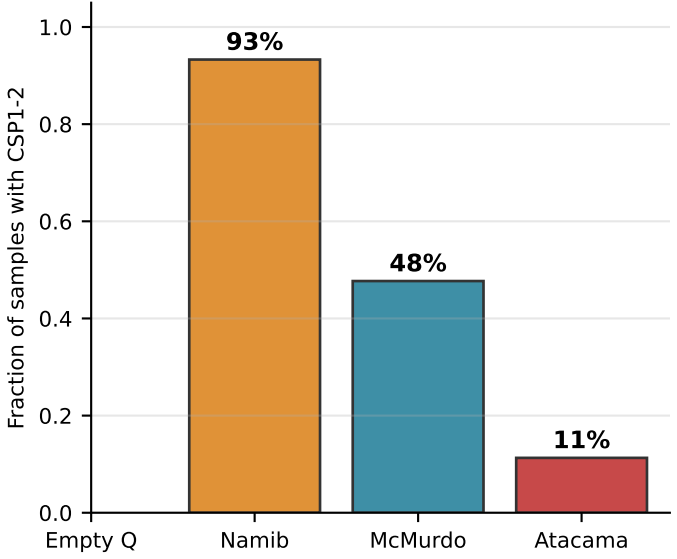
(b) Deep — CSP1-2 ego network (14 co-occurring genera)



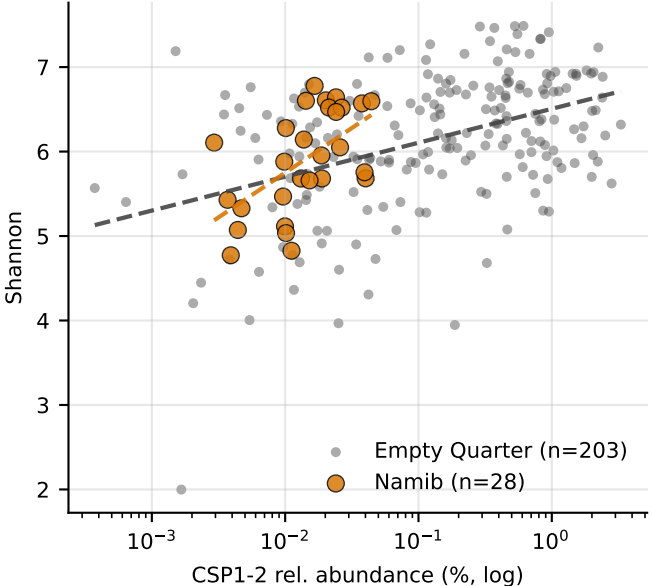
(c) Rhizosphere — CSP1-2 ego network (14 co-occurring genera)



(d) Cross-desert prevalence of CSP1-2 at 85% V4 ID



(e) CSP1-2 rel. abundance vs Shannon (EQ + Namib)
EQ: $\rho=+0.37$, $p=4.4\text{e-}08$ | Namib: $\rho=+0.58$, $p=1.1\text{e-}03$



(f) Salinity vs Shannon (EQ + Atacama)
EQ (XRF S): $\rho=-0.34$, $p=1.4\text{e-}06$ | Atacama (EC): $\rho=-0.38$, $p=1.2\text{e-}02$

